

TECHNICAL DESCRIPTION

ZJU-DDRX Autonomous Racing Vehicle — Competition Submission Edition

Team	ZJU-DDRX	Event	29th ROBORACER Grand Prix
Submission	Hardware and software summary	Prepared	30 July 2026
Vehicle	Single-motor autonomous race car	Scope	Competition-required disclosure

Purpose. This document provides the competition-required overview of the vehicle’s hardware and software technologies. It intentionally excludes tuning values, algorithmic implementation details, calibration data, and performance limits.

1. System Overview

The ZJU-DDRX vehicle is a self-contained autonomous racing platform. A single Intel NUC onboard computer receives measurements from one Livox Mid-360 LiDAR, executes the autonomy pipeline, and issues steering and propulsion commands. Steering is provided by a dedicated digital servo. Propulsion uses one brushless motor driven through a VESC-based electronic speed controller. All driving computation is performed onboard.

Functional signal path

Livox Mid-360 sensing	Point-cloud and vehicle-state inputs	Intel NUC autonomy stack	Steering + speed commands	6501 servo + VESC actuation
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2. Vehicular Hardware Stack

Subsystem	Model / implementation	Role in the vehicle
Chassis	Serpent SRX8GTE R 1/8 4WD EP chassis (SER600074)	Mechanical platform, suspension, steering linkage, and four-wheel drivetrain.
Main computer	Intel NUC onboard computer	Single vehicle computing unit for sensing, autonomy, control, logging, and supervision.
LiDAR	Livox Mid-360	Single 3D LiDAR used as the primary exteroceptive sensor for environment and track perception.
Steering actuator	6501 full-metal digital servo (vendor-listed model)	Converts steering commands into front-wheel steering actuation.
Drive motor	Trackstar 4268 2200KV brushless motor	Single propulsion motor coupled to the chassis drivetrain.

Subsystem	Model / implementation	Role in the vehicle
Motor controller	VESC-based electronic speed controller	Receives propulsion commands, controls the motor, and provides the electrical drive interface.

The vehicle uses one main computing unit, one LiDAR, and one propulsion motor. Final assembled dimensions, the detection enclosure, bumper, and battery arrangement will be presented for on-site inspection in accordance with the rules.

3. Vehicular Software Stack

The race software is organized as a modular onboard stack. Modules exchange time-stamped sensor, state, trajectory, and actuator-command messages. The summary below describes responsibilities and interfaces only; it does not disclose algorithm choices, tuning, thresholds, or internal data products.

Layer	Primary functions	Interface / output
Platform and runtime	Onboard runtime on the Intel NUC; process launch, configuration loading, health supervision, and onboard recording.	Provides a repeatable runtime for all vehicle functions.
Device interfaces	Mid-360 acquisition, VESC communication, steering-servo command output, and vehicle-status inputs.	Normalizes hardware data and actuator commands for the rest of the stack.
Perception	LiDAR point-cloud conditioning and extraction of track- and obstacle-relevant information.	Produces environment observations used by localization and motion planning.
Localization and state estimation	Maintains an onboard estimate of vehicle pose and motion using available sensor and map information.	Supplies the common vehicle state consumed by planning and control.
Planning	Represents the drivable track, evaluates the local racing situation, and generates a feasible motion reference.	Outputs a trajectory or equivalent reference without requiring offboard computation.
Control	Closed-loop steering and longitudinal command generation with command limiting and actuator conversion.	Translates the motion reference into servo and VESC commands.
Safety and operations	Toggle emergency-stop handling, state checks, fault response, start/restart handling, diagnostics, and logging.	Supports inspection, safe operation, and post-run review.

4. Integration and Data Flow

At startup, the onboard supervisor initializes the sensor and actuator interfaces, then brings up the autonomy modules. LiDAR and vehicle-status data are time-stamped and processed onboard. The localization and state-estimation layer provides a consistent vehicle state. The planning layer produces a motion reference, and the control layer converts that reference into bounded steering and propulsion commands.

During normal driving, the autonomy loop does not depend on external computation or incoming operational data. A laptop may be used only in the rule-permitted visualization or debugging role. Operational records are stored onboard for diagnosis and verification.

5. Safety and Competition Alignment

Topic	Team implementation statement	Submission relevance
Onboard autonomy	All perception, state estimation, planning, and control computation runs on the vehicle's Intel NUC.	Aligned with the onboard-computation requirement.
Compute / LiDAR / motor count	One main computing unit, one Livox Mid-360 LiDAR, and one Trackstar propulsion motor.	Aligned with the permitted component counts.
Emergency stop	A toggle-type emergency-stop function is included in the operational safety layer and will be demonstrated at inspection.	Supports the software-inspection requirement.
External communication	No data is transmitted to the vehicle during normal autonomous driving; any external laptop is limited to permitted visualization/debugging use.	Aligned with vehicle computation and communication restrictions.
Physical inspection items	Final chassis envelope, soft bumper, detection box, battery cell count, and mechanical security will be checked before competition operation.	Prepared for on-site hardware inspection.

6. Software Availability and Disclosure Scope

The team maintains the race-stack project at the repository address below. This submission is an architectural description for registration and inspection. Source-level implementation, maps, calibration files, control gains, model parameters, and performance envelopes are not included because they are not required to understand the vehicle's hardware and software stack.

Repository: <https://github.com/ZJU-DDRX/ddrx-race-stack>

References

- 29th ROBORACER Grand Prix Rules (last updated 15 July 2026): <https://2026ifac-robocer.com/en/rules.html>
- Team hardware budget workbook supplied for this submission, used only to verify the chassis, motor, and steering-servo model names.